

HOSSAIN AHMMED TAUFIQ

Software Engineer · Backend · AI/ML Systems

Dhaka, Bangladesh · hossainahmedtaufiq22@gmail.com · +880 1728-360834

[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

Software Engineer with experience building scalable full-stack web applications, backend services, and AI-powered products. Experienced in React, Next.js, Node.js, Express, TypeScript, PostgreSQL, MongoDB, and modern LLM integrations. Founder of Softlligence Technologies and contributor to production software used by businesses and educational institutions. Currently pursuing a B.Sc. in Computer Science while conducting research in Multimodal RAG and Deep Learning.

TECHNICAL SKILLS

Programming Languages: TypeScript, Python, JavaScript, Java, C++, SQL

Frontend: React.js, Next.js, React Native, Tailwind CSS, HTML5, CSS3, Redux, Framer Motion

Backend: Node.js, Express.js, Django, REST APIs, GraphQL, JWT, Microservices, WebSockets

AI / ML: LLM Integration, RAG Systems, LangChain, Scikit-learn, TensorFlow, Gemini API, Prompt Engineering

Databases: PostgreSQL, MySQL, MongoDB, Firebase

Cloud & DevOps: Git, GitHub, GitHub Actions, Docker, AWS, Vercel

Methodologies: Agile, CI/CD, Testing, OOP, DSA

PROFESSIONAL EXPERIENCE

Software Engineer · MEV (Mango Electric Vehicle) · Electric Vehicle Manufacturer

Jan 2026 -Present

On-site

- Maintain and enhance MEV's internal management systems, supporting day-to-day operations across the business.
- Develop applications for MEV's locally built electric vehicles, delivering software for the in-house vehicle platform.
- Build and maintain CRM and customer-facing tools using Next.js, Express.js, PostgreSQL, and TypeScript.

Full-Stack Developer · Brooksource · Enterprise React Consultancy

Apr 2023 – Sep 2024

Remote · US-based staffing firm delivering React solutions to Fortune 500 clients

- Developed and maintained production React.js + TypeScript applications with Redux serving high-traffic enterprise clients.
- Reduced front-end load time by 20% through code splitting, lazy loading, and bundle optimisation; designed API integration layers that measurably reduced cross-service latency.
- Consistently shipped features ahead of schedule across distributed Agile teams spanning multiple time zones.

Associate, Web Developer · Americares

Feb 2023 – Nov 2023

- Built and maintained web applications as an Associate Web Developer, exploring a wide range of tools and practices to grow quickly as an engineer.

SELECTED PROJECTS

Softlligence Manufacturing Cloud- [Enterprise Manufacturing ERP/MIS SaaS] · Built a multi-tenant Manufacturing ERP/MIS platform for industrial businesses with modular architecture and enterprise-grade scalability.

[Live URL](#) · [GitHub](#)

- Developed core ERP modules including authentication, multi-tenancy, RBAC, inventory, manufacturing, commercial operations, and reporting using **AWS, Next.js, Node.js, Express, TailwindCSS, Prisma, PostgreSQL and TypeScript**.
- Designed a reusable industry-template framework supporting multiple manufacturing sectors while ensuring secure tenant isolation, scalable architecture, and AI-ready platform integration.

Playpen School Website [Ongoing] · Payment Gateway · Next.js · TypeScript · Node.js · Tailwind CSS · PostgreSQL · Express.js · AWS

[Live URL \(Current Production\)](#) · [Personal Preview](#) · [GitHub](#)

- Developed a production-ready school management platform enabling online admissions, tuition payments, event management, and parent engagement through an intuitive web interface.

- Built reusable UI components and secure backend services with TypeScript and Express.js, improving maintainability, scalability, and future feature development.

Mango Electric Vehicle Website · Node.js · Tailwind CSS · PostgreSQL · Express.js · Next.js · TypeScript · AWS · Framer Motion

[Live URL](#) · [Private Repo](#)

- Engineered a scalable full-stack platform using Next.js, Express.js, and PostgreSQL, implementing secure reservation workflows, inquiry management, payment integration, and role-based backend APIs.
- Optimized Core Web Vitals through server-side rendering, image optimization, caching strategies, and SEO best practices, delivering a fast, responsive, and accessible user experience.

FocusFlow –Desktop App · Python · PySide6 · Matplotlib

[GitHub](#)

- Architected a fully offline productivity application using Python and PySide6, integrating task management, habit tracking, calendar scheduling, note-taking, analytics, and automated backup systems
- Designed a modular desktop architecture with local data persistence and interactive data visualization, providing a fast, privacy-focused productivity solution without cloud dependencies.

RESEARCH

Multimodal RAG with Cross-Modal Hallucination Reduction · North South University

Aug 2025 – Present

Supervisor: Dr. Nabil Bin Hannan · Preparing for conference submission

- Designed and implemented a multimodal Retrieval-Augmented Generation (RAG) framework integrating text, images, and tables, applying cross-modal hallucination reduction techniques to improve factual consistency and reliability in LLM-generated responses.

Multi-Task ML for Band Gap & Formation Energy Prediction · North South University

Sept 2025 – Jan 2026

[Publication](#)

- Developed a multi-task deep learning model for simultaneous band gap classification and formation energy prediction, achieving higher predictive performance and computational efficiency compared to single-task baseline models.

Validating Stock Market Price Predicting Algorithms · North South University

Mar 2026 – Present

[Publication](#) · [GitHub](#)

- Built an end-to-end machine learning pipeline for stock price forecasting using financial time-series and news sentiment analysis, evaluating ARIMA, LSTM, Random Forest, XGBoost, and LightGBM models, with Random Forest achieving the best predictive accuracy.

EDUCATION

B.Sc. Computer Science & Engineering · North South University, Dhaka

Expected Dec 2026

- CGPA: 3.83/ 4.00 · Relevant Coursework: DSA, Machine Learning, AI, Software Engineering, Database Systems, Web Development
- Research: Epilepsy Detection (Deep Learning) · Multimodal RAG (NLP/CV)

HSC, Science · Notre Dame College, Dhaka

Jan 2020

- GPA: 5.00 / 5.00
- Volunteered in organizing and hosting college events.
- Developed teamwork, leadership, public speaking, and event management skills.

CERTIFICATIONS & LANGUAGES

Certifications: Full-Stack Web Dev (Programming Hero) · DSA (Udemy & Phitron) · AI & ML (GP Academy, Programming Hero, Udemy) · Public Speaking (BYLC) · Bash Scripting (freeCodeCamp) · Arduino (Gobeshona)

Languages: Bangla — Native · English — Professional Fluency (written & spoken)